Bita Karvizi

bita.karvizi@student.kit.edu

+49(0) 1743639399 | Personal Website

Github | Linkedin

SUMMARY

I am Bita Karvizi, a Master's student at the Karlsruhe Institute of Technology (KIT) and an Android Developer with over four years of professional experience. I have developed multiple production-grade applications using Kotlin and gained experience in building secure and hardware-connected applications. Alongside my professional experience, I have an academic background in software engineering and research experience from my bachelor's studies and subsequent work as a Research Assistant.
I am interested in applying sound software engineering principles and combining practical development experience with research-driven problem-solving to build reliable and maintainable software systems.

SKILLS

Technical

- **Android Development**: API and Database Integration, Jetpack Libraries (Compose, Flow, Coroutines), Dependency Injection, Clean Modular Architecture, Real-time Data Handling
- **Application Security**: JNI using C++, Symmetric and Asymmetric Encryption, Secure Local Storage
- **Programming Languages**: Kotlin, Java, SQL, Python, C, C++, JavaScript, HTML/CSS

PROFESSIONAL EXPERIENCE

Android Developer at SeeNous (Website) Jul. 2025 – Feb. 2026

- Developed an Android application for real-time monitoring of an ESP32-based vibration sensor using **custom MQTT/UDP communication**.
- Implemented **digital signal processing** (DSP) methods and real-time time-series visualization for efficient signal analysis.
- Optimized latency and performance for heavy real-time charts in a ViewPager, ensuring smooth rendering, memory efficiency, and stable UI under continuous high-frequency updates.

Android Developer at Part Software Group (Website) Nov 2021 - Feb 2024

- **Projects**
  - Farashenasa Application (Myket - Bazar)
    * Developed an application to manage digital signature certificates, with over 2,000 active users.
    * Designed and implemented **a**uthentication, **messaging** modules, and an integrated PDF viewer for seamless document handling.
    * Ensured application security by implementing encryption protocols and securing the app with JNI.
  - Demo Version of Signal Application (Myket -Bazar)
    * Designed and developed a demo version of the Signal app to track financial market trends.
    * Implemented MVVM and Clean Architecture for scalable, maintainable code.
    * Implemented interactive charts powered by API data.

Android Developer at FAVA Center of FUM Apr 2024 - Feb 2025

- **Projects**
  - University Portal Application: Developed the official new version of the university's student portal.
  - Employee Attendance Application: Developed an application for tracking employee attendance using RFID to scan ID cards.

Android Development Intern at Part Financial Data Processing CO. Jun 2021 - Nov 2021
- Final Project: Coin Ranking (Github)
- Developed a real-world application from scratch practicing different concepts of Android.

ACADEMIC EXPERIENCE

Research Assistant at Software Quality Lab (Website) Apr 2024 - Apr 2026
- Supervised by Dr. Abbas Rasoolzadegan
- Explored **automated testing** frameworks and techniques from scholarly research.
- Gained expertise in **automated test repair** methods for evolving software versions.

Teaching Assistant
- Fundamentals of Programming (C Programming) - Dr. Saeid Abrishami Fall 2024
- Algorithm Design - Dr. Mostafa Nouri-Baygi Fall 2024
- Finite Automata and Languages Dr. Saeid Abrishami Spring 2024
- Fundamentals of Programming (C Programming) - Dr. Saeid Abrishami Fall 2023
- Fundamentals of Programming (C Programming) - Dr. Mostafa Nouri-Baygi Fall 2022
- Computer Systems Architecture - Dr. Sara Ershadi-Nasab Fall 2021, Spring 2022

EDUCATION

Karlsruhe Institute of Technology (KIT) | *Master of Science in Computer Science* **2026 - Present**

Ferdowsi University of Mashhad | *Bachelor of Computer Engineering* **2019 - 2024**
- **GPA**: 3.55/4 (17.18/20)
- **Bachelor's Project**: Android Application for Student Attendance Management (Github)

OPEN-SOURCE PROJECTS

IMDb Movies Android Application | *GitHub* **2025**
- Developed a modern Android application using Jetpack Compose and MVI architecture.
- Implemented Hilt for dependency injection, Retrofit for API communication, and Coroutines with StateFlow for asynchronous state management.
- Utilized Compose Navigation and LaunchedEffect for screen navigation and UI side-effect handling.

Presence and Absence Android Application | *GitHub* **2024**
- Developed an exam attendance management app using the latest Jetpack libraries, Hilt, and Material Design.
- Integrated security features such as token handling, root detection, and code obfuscation.

Coin Ranking Android Application | *GitHub* **2022**
- Developed a cryptocurrency exploration app using MVVM, Room, Retrofit, Coroutines, and Flow, featuring search, bookmarking, and light/dark themes.

PBFT Consensus Simulation in NS3 | *GitHub* **2024**
- Implemented the Practical Byzantine Fault Tolerance (PBFT) consensus algorithm in C++ on the NS3 network simulator, as a team final project for a Blockchain course.

Network-Based Game List using JavaFx | *GitHub* **2020**
- Developed a Java/JavaFX desktop app with a networked architecture, using WebSockets for real-time chat and communication with a local Java backend.
- Integrated SQL database support for managing users, game data, and chat logs, with a clean modular design across UI, networking, and data layers.

LANGUAGE

English IELTS: 7.5 (R8.5, L8, S6.5, W6.5)

Persian Native